

August 2026

# THELOOK ECOMMERCE

Customer Analysis & Segmentation

Nguyen Thi Van Anh

Github: [https://github.com/oopsvananh/thelook\\_project](https://github.com/oopsvananh/thelook_project)

# OUR CONTENTS

01. Project Overview

02. Dataset Overview

03. Terms & Metrics

04. Approach

05. Performance Overview

06. Executive Summary

07. Recommendation

08. Detailed Analysis

# PROJECT OVERVIEW

- TheLook is a fashion eCommerce platform operating from 2019 to 2024
- This project focuses on customer analysis: understanding who customers are, how much value they generate, and how their behaviors differ across segments.
- The goal is to go beyond “what happened” to uncover where the growth opportunities are and what TheLook should do next.

***How can TheLook grow revenue by better understanding its customer segments and their product preferences?***



# DATASET OVERVIEW

Table users: Users' identity & demographics

| Column         | Meaning                                       |
|----------------|-----------------------------------------------|
| id             | Unique identifier for each user               |
| age            | Age of the user                               |
| gender         | Gender of the user                            |
| country        | Country of residence for the user             |
| traffic_source | Source from which the user arived at the site |

*Note: Only columns relevant to this analysis are shown*

Table orders: Order-level information

| Column      | Meaning                                                           |
|-------------|-------------------------------------------------------------------|
| order_id    | Unique identifier for each order                                  |
| user_id     | Identifier for the user who placed the order                      |
| status      | Order status (Complete, Cancelled, Processing, Shipped, Returned) |
| created_at  | Timestamp when the order was created                              |
| num_of_item | Total number of items in the order                                |

# DATASET OVERVIEW

Table order\_items: Product-level detail within each order

| Column     | Meaning                                 |
|------------|-----------------------------------------|
| id         | Unique identifier for each order item   |
| order_id   | Identifier for the associated order     |
| product_id | Identifier for the product in the order |
| sale_price | Sale price of the order item            |

Table products: Product information

| Column     | Meaning                                     |
|------------|---------------------------------------------|
| id         | Unique identifier for each product          |
| category   | Category to which the product belongs       |
| cost       | Cost price of the product                   |
| department | Department where the product is categorized |

*Note: Only columns relevant to this analysis are shown*



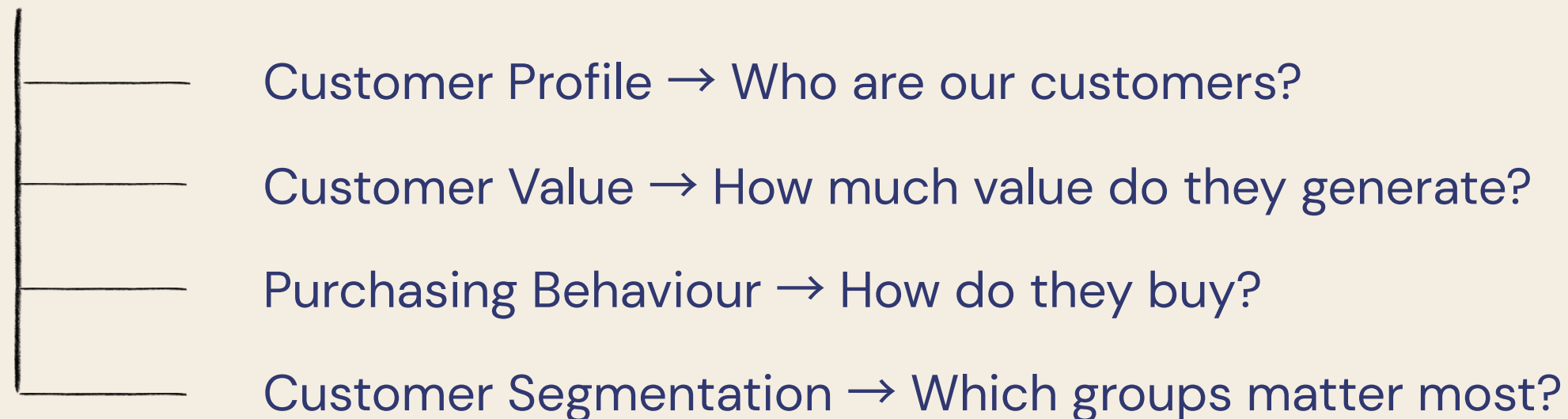
# TERMS & METRICS

| Term                      | Definition                                                                                                  |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| AOV (Average Order Value) | Average revenue per completed order (Total Revenue/Completed Orders)                                        |
| Active Customers          | Registered users with at least one valid order, excluding Cancelled and Returned.                           |
| Completed Customers       | Active customers with at least one order in "Complete" status                                               |
| Active Orders             | Orders with status Shipped, Processing, or Complete (excludes Cancelled and Returned)                       |
| RFM                       | Recency (days since last purchase) – Frequency (number of completed orders) – Monetary (total amount spent) |

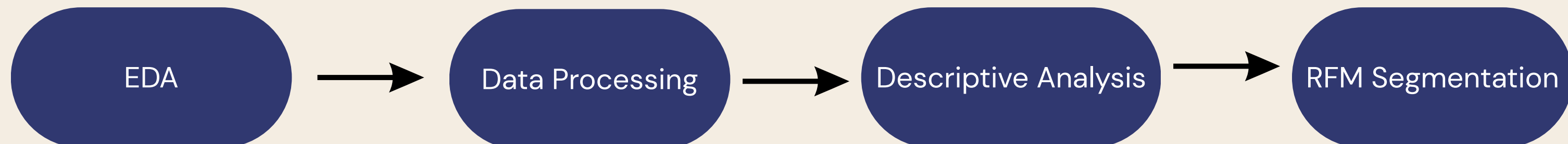
# APPROACH

## 1. Analysis Framework

How can TheLook grow revenue ?



## 2. Methodology



# PERFORMANCE OVERVIEW

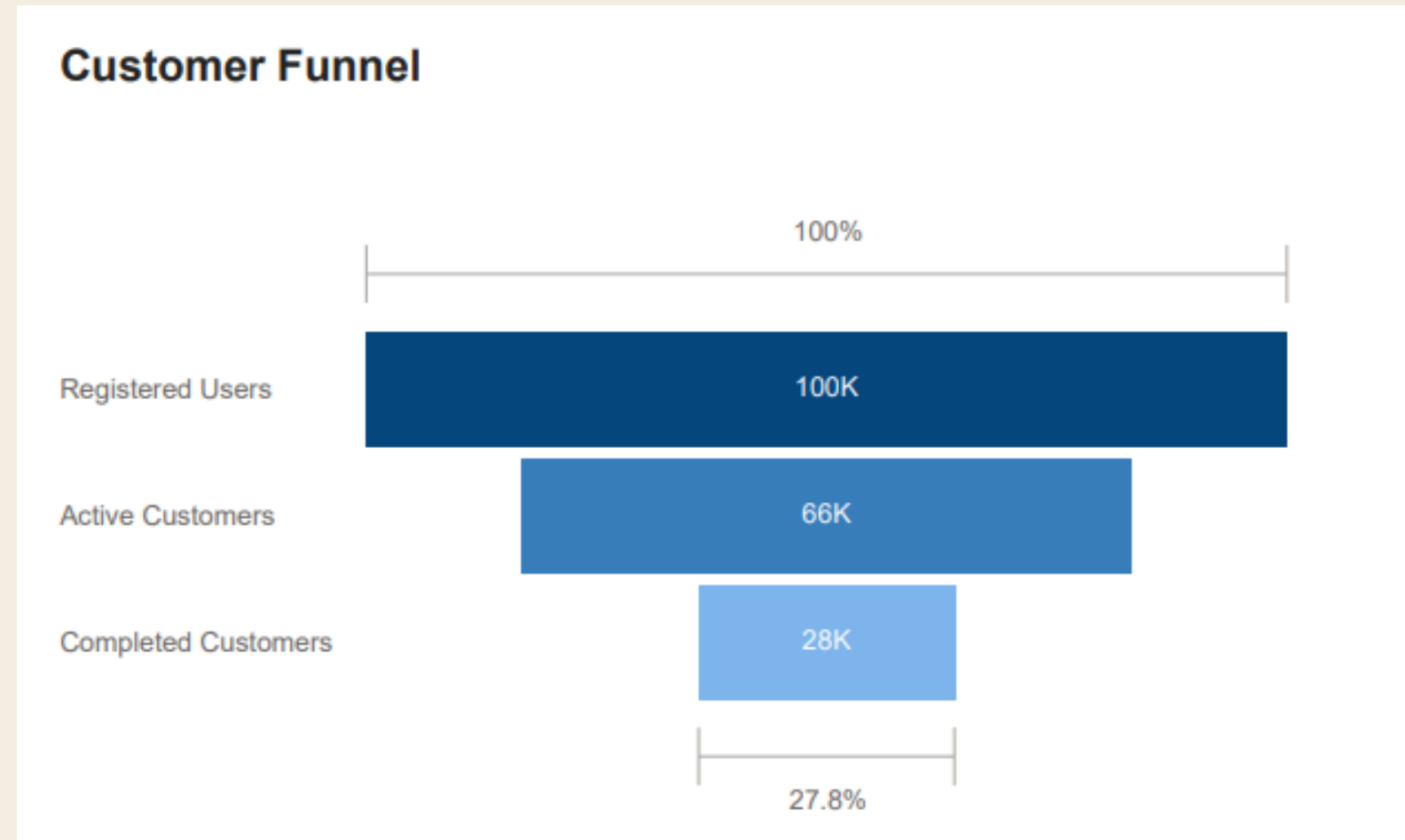
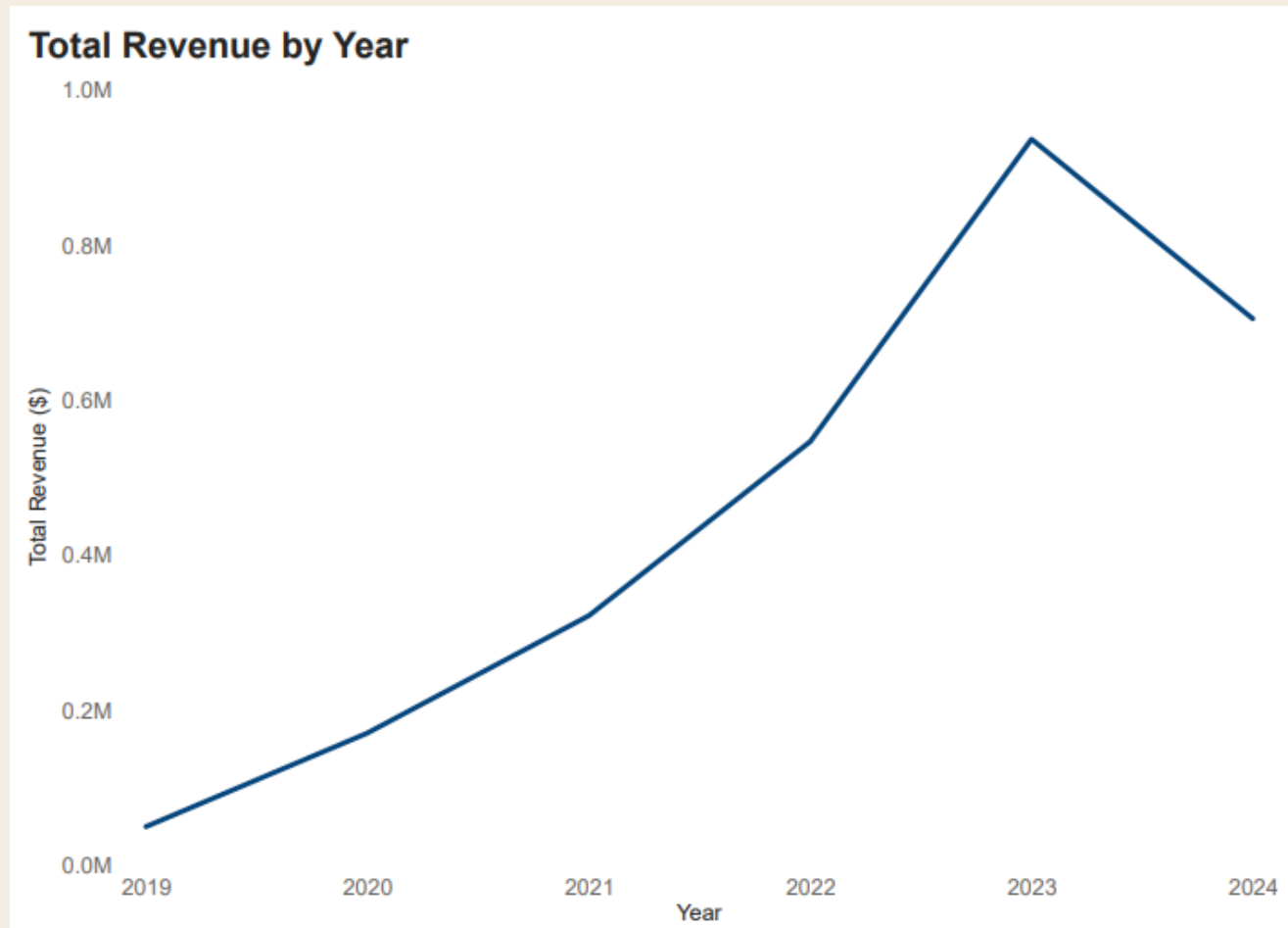
2.73M  
Total Revenue

1.41M  
Profit

\$86.84  
AOV

100K  
Total Users

94K  
Active Orders



Note: 2024 reflects partial-year data and is not directly comparable to prior full years.

***TheLook shows healthy scale and margins, but a large share of users never convert into completed revenue***



# EXECUTIVE SUMMARY

*Low repeat purchasing highlights an opportunity to increase customer value through retention.*

## **1. A large user base, but significant drop-off before completed purchase**

100K registered users → 66K active customers → only 27.8K completed customers (27.8%).

## **2. Behavior matters more than customer profile**

Customer demographics show limited differentiation, while acquisition is highly concentrated in Search; neither clearly explains customer value.

## **3. Repeat purchasing is limited in the observed data.**

88.09% of customers purchased only once, while only 10.74% purchased twice and 1.18% purchased three or more times. Even the 2019 cohort retained only around 5.5% after one year.

## **4. High-value segments contribute disproportionately to revenue**

Loyal Customers represent only 1.17% of customers, while High-Value New Customers represent 4.37%. High-Value New Customers have an AOV of \$219.89, while Loyal Customers average 3.06 orders per customer.

# RECOMMENDATION

**Key message: "Prioritize second and subsequent purchases through targeted engagement based on customer purchase history."**

## **1. Convert one-time buyers into repeat buyers**

Increase second-purchase conversion through timely post-purchase engagement:

- Trigger email/push 7–14 days after first purchase
- Win back Inactive/Lost via recency, not demographics.

## **2. Develop Repeat Buyers Toward Loyal**

Use frequency-based incentives to move Repeat Buyers (2.00 orders/customer) closer to Loyal behavior (3.06 orders/customer):

- Loyalty program rewarding purchase count, not spend
- Recommend categories beyond the core 3 to accelerate diversification

# RECOMMENDATION

**Key message: "Prioritize second and subsequent purchases through targeted engagement based on customer purchase history."**

## 3. Retain High-Value New

Encourage High-Value New customers to make a second purchase while reinforcing their high-value shopping behavior:

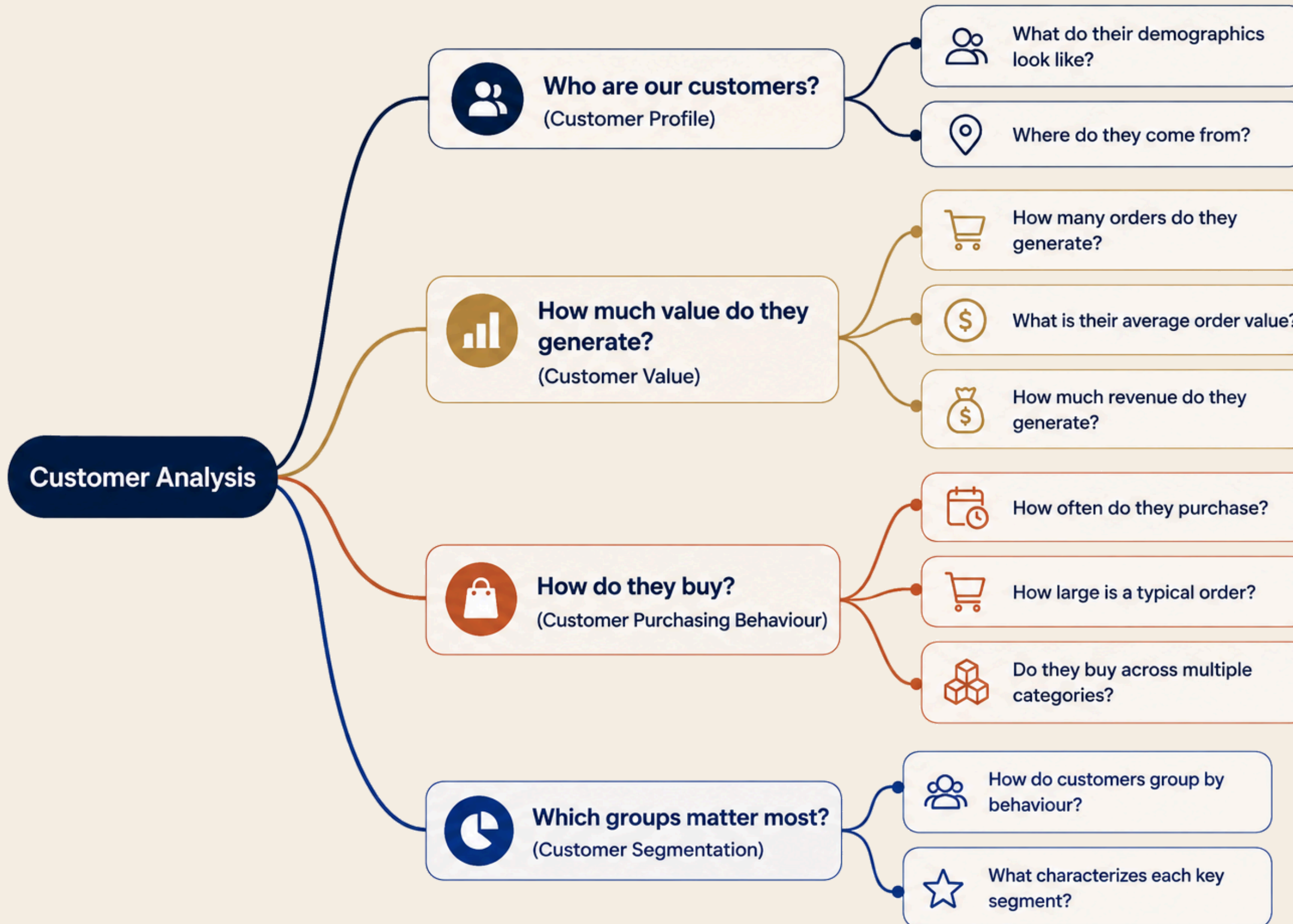
- Recommend products from their preferred categories.
- Use personalized cross-selling to extend their first-purchase value.

## 4. Deepen Engagement for Loyal Customers

Loyal Customers have lower top-3 category revenue concentration than Repeat Buyers (28.18% vs. 31.17%). Test personalized recommendations to encourage further purchases:

- Recommend complementary products based on each customer's purchase history.
- Test familiar-category recommendations against new-category suggestions for Loyal Customers and Repeat Buyers separately.
- Measure incremental repeat purchases and revenue per customer before expanding the approach.

# DETAILED ANALYSIS





# CUSTOMER PROFILE

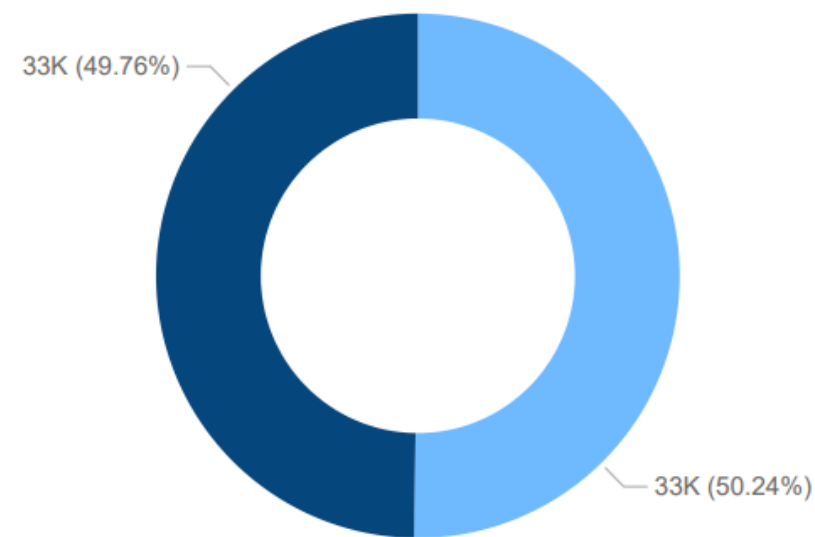
*Search accounts for most active customers*

## Takeaways:

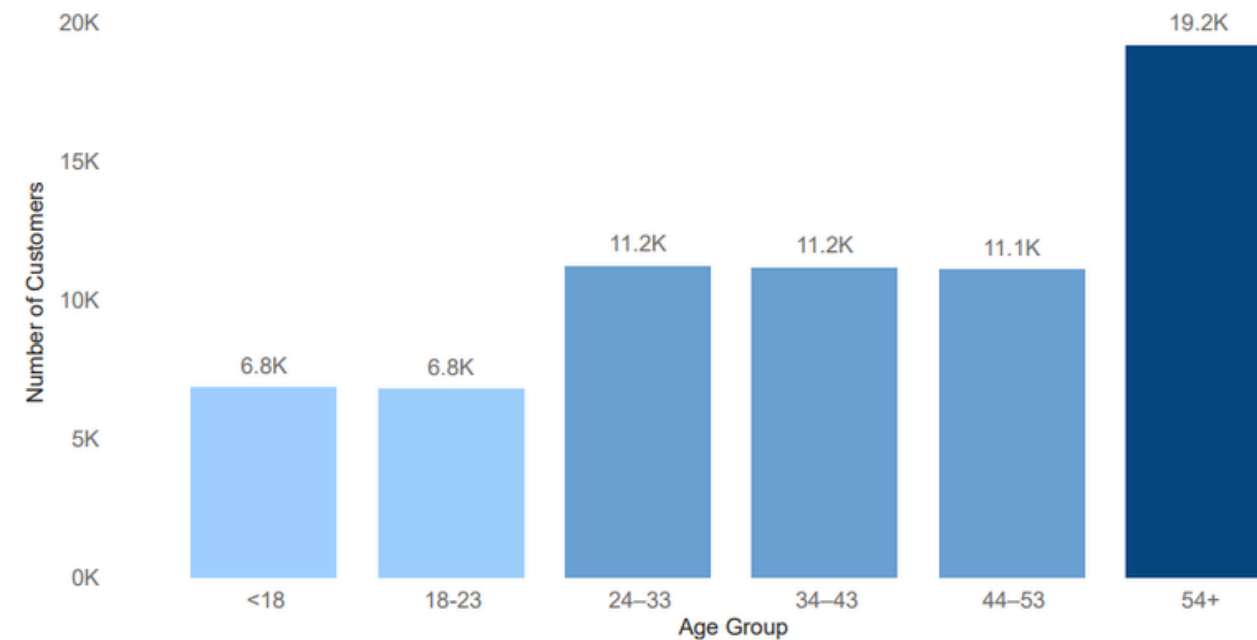
Customer demographics are broadly balanced, while acquisition is concentrated in Search – neither clearly explains customer value.

1. Gender is nearly evenly split – 50.24% Male vs. 49.76% Female.
2. Age skews toward 54+ (19.2K, ~29%), but this open-ended bracket covers a wider age range than the other groups.
3. China (22.5K) and the US (14.7K) have the largest customer bases, but higher customer volume does not imply higher value.
4. Search dominates acquisition (~5× Organic), but order-status mix is broadly similar across channels, suggesting volume rather than customer quality drives the difference.

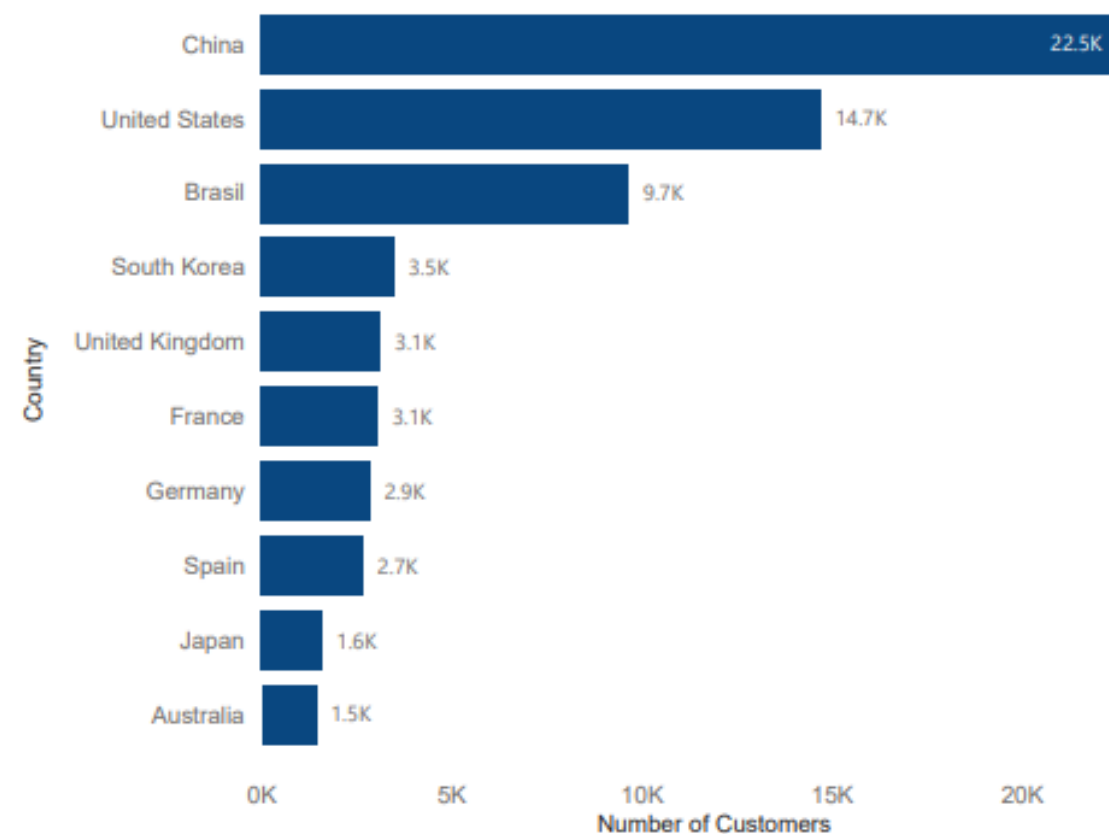
Active Customers by Gender



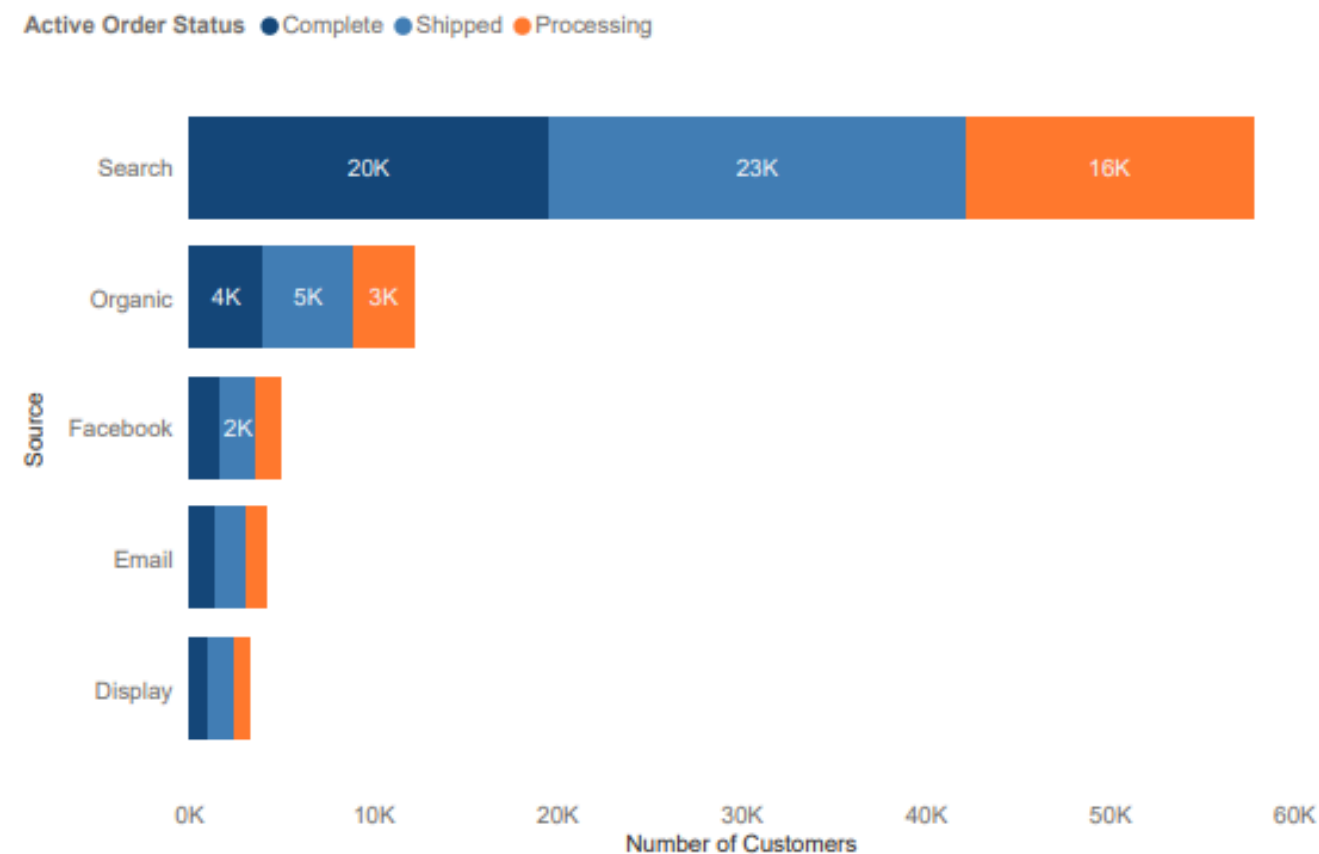
Active Customers by Age Group



Active Customers by Country



Active Customers by Traffic Source



# CUSTOMER VALUE

*45.66% of completed customers generate 80% of revenue*

66K

Active Customers

27.78%

Completed Customer Rate

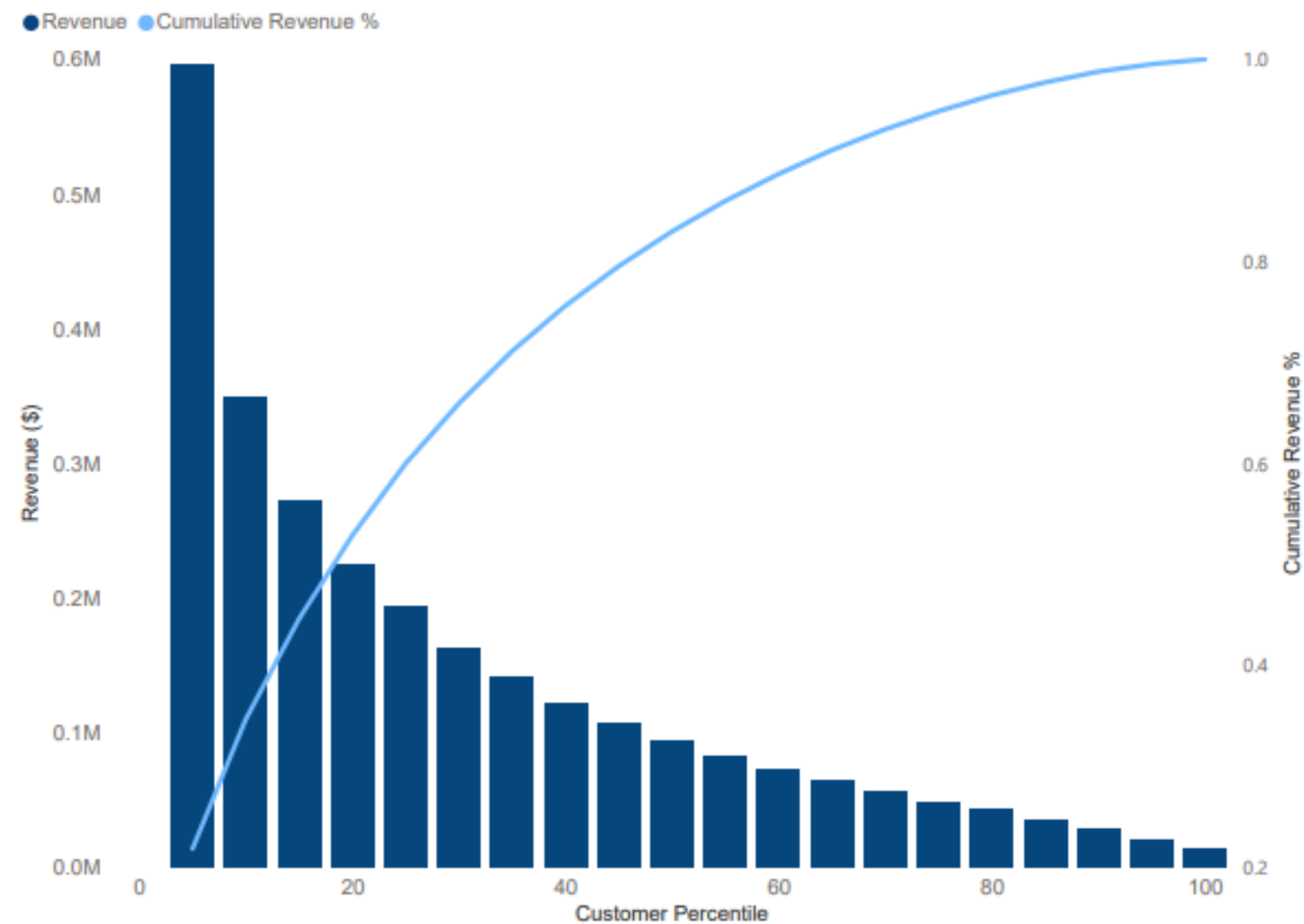
\$98.27

Revenue per Completed  
Customer

11.91%

Repeat Purchase Rate

Customer Revenue Concentration (Pareto)



## Takeaways:

Customer value is limited by low conversion and weak repeat purchasing – most customers generate value only once.

1. Only 27.78% of registered users become completed customers, leaving a large pool of customers with no realized revenue
2. Repeat purchase rate is only 11.91%, meaning most converted customers do not return to purchase again
3. Revenue concentration is moderate, not extreme – roughly 45% of customers generate 80% of total revenue, meaning value is spread across a fairly broad base rather than concentrated in a small VIP group

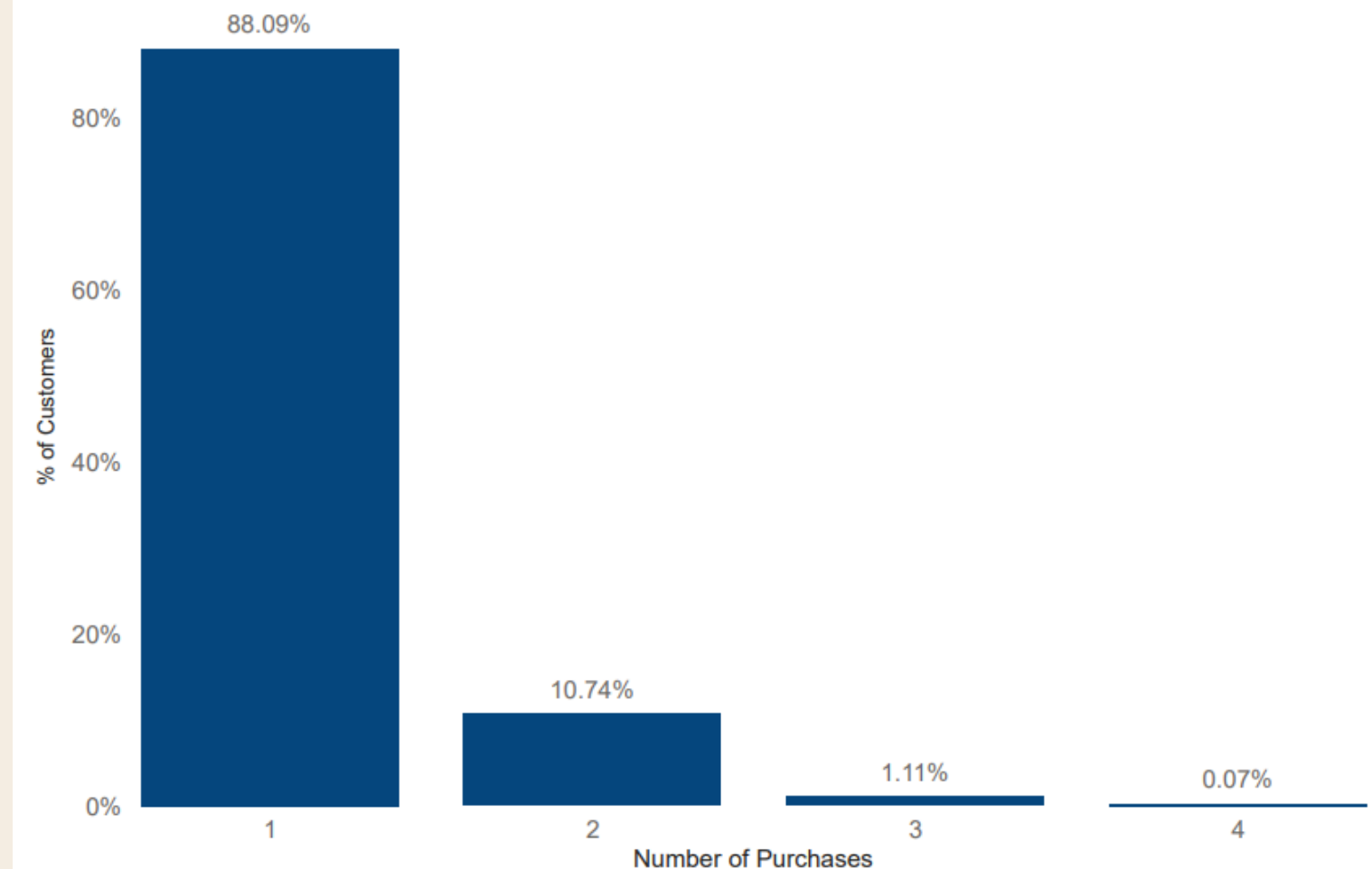
# PURCHASING BEHAVIOUR

*88.09% of completed customers have only one completed order*

## Takeaways:

1. 88.09% of customers purchase only once, while just 10.74% purchase twice and only 1.18% purchase three or more times
2. Even the 2019 cohort – with 5 full years to return – only retained ~5.5% of customers by year 1, confirming this pattern isn't due to limited observation time, but reflects a real, structural retention gap

Distribution of Customers by Purchase Frequency



Customer Retention Rate by Yearly Cohort

| cohort_year | 0      | 1    | 2    | 3    | 4    | 5    |
|-------------|--------|------|------|------|------|------|
| 2019        | 100.00 | 5.49 | 6.02 | 4.60 | 3.01 | 1.42 |
| 2020        | 100.00 | 5.40 | 4.75 | 4.64 | 1.73 |      |
| 2021        | 100.00 | 6.32 | 6.38 | 2.73 |      |      |
| 2022        | 100.00 | 8.27 | 3.19 |      |      |      |
| 2023        | 100.00 | 5.14 |      |      |      |      |
| 2024        | 100.00 |      |      |      |      |      |

# PURCHASING BEHAVIOUR

*Completed orders average 1.45 items and \$86.84, with  
Jeans reaching the most customers*

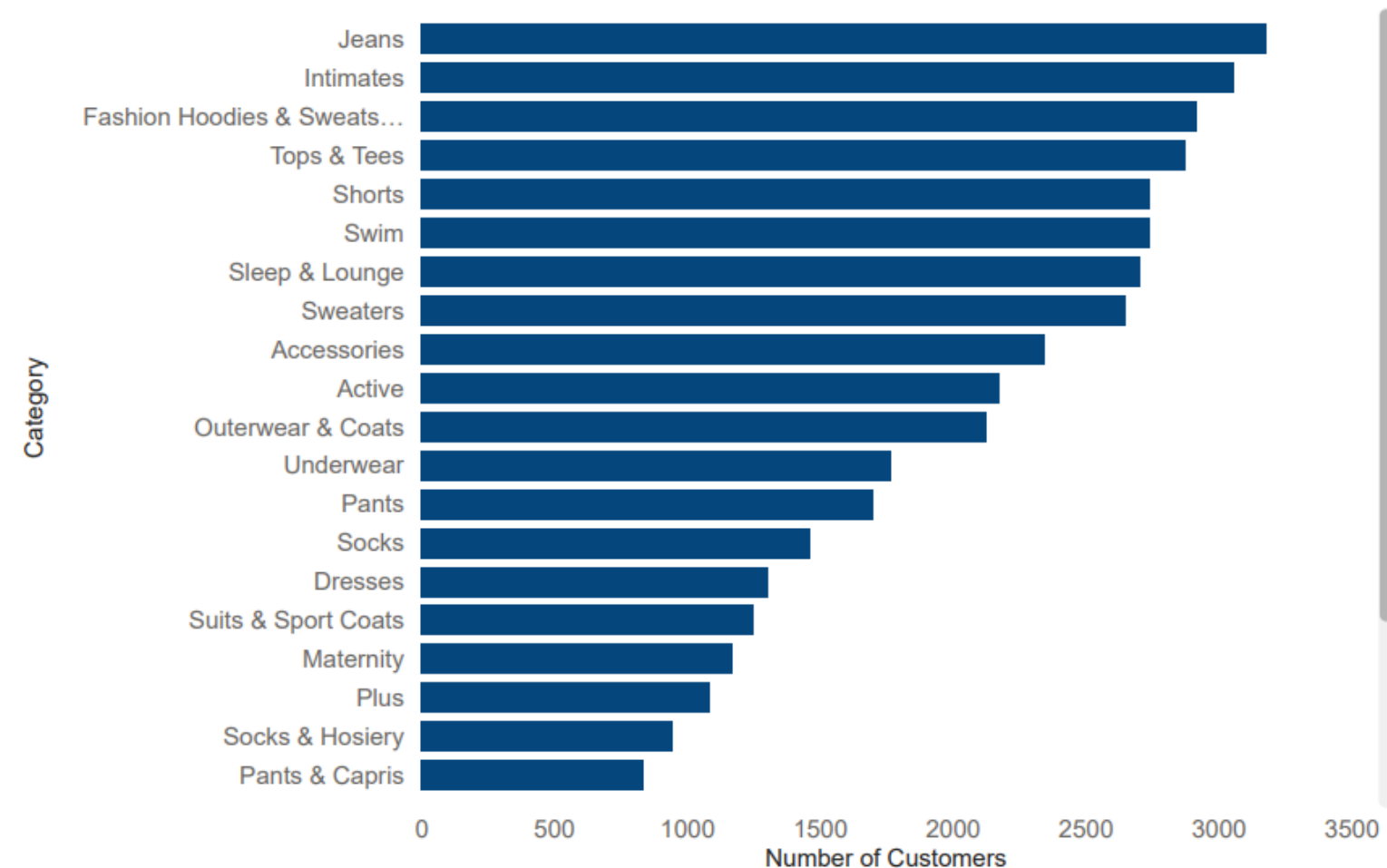
## Takeaways:

1. Basket size is small overall – Completed orders average 1.45 items and \$86.84, leaving room to grow order value through cross-selling or bundling.
2. Jeans reaches the most completed customers, followed by Intimates and Fashion Hoodies & Sweatshirts. These categories provide starting points for testing product recommendations.

1.45  
Items per order

\$86.84  
AOV

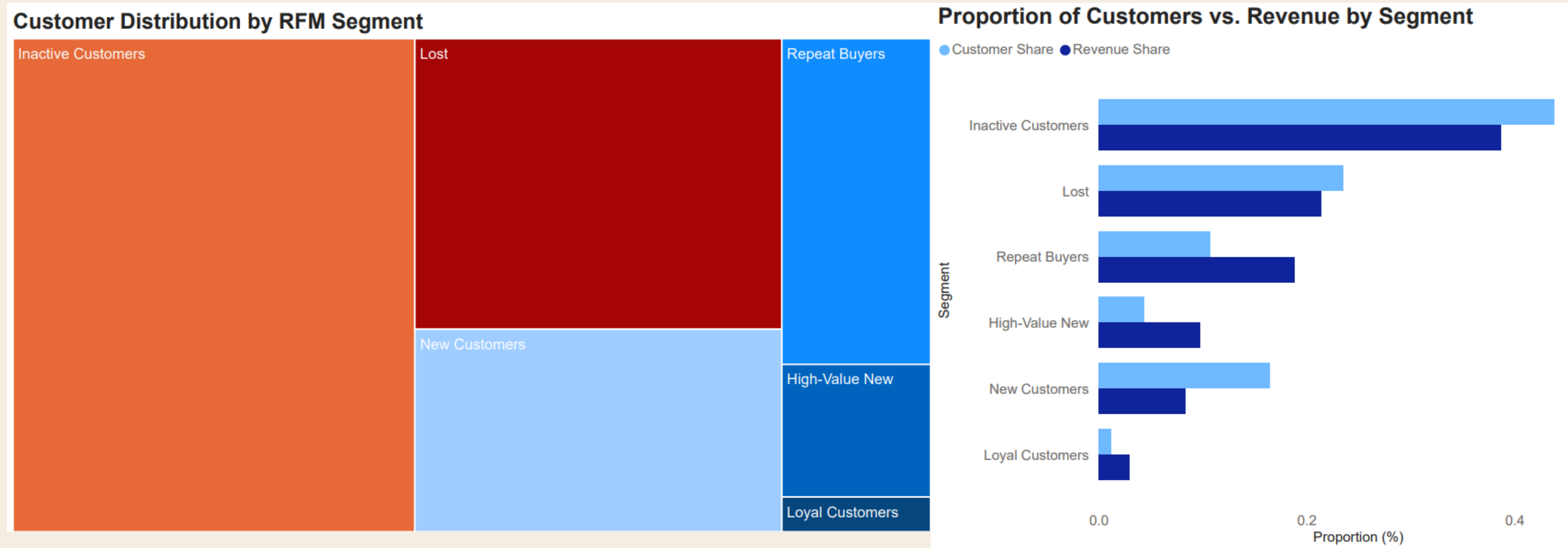
Customers by Product Category





# CUSTOMER SEGMENTATION

*Inactive and Lost segments account for 67.26% of completed customers*



## Takeaways:

1. Inactive + Lost customers account for 67.26% of the base – the largest group by far, but with limited ongoing value
2. Loyal Customers represent just 1.17% and High-Value New just 4.37% – the two smallest segments, but the most valuable
3. Customer share and revenue share diverge sharply by segment – Inactive/Lost hold the largest customer share but a disproportionately smaller revenue share, while Loyal/High-Value New show the opposite pattern

# CUSTOMER SEGMENTATION

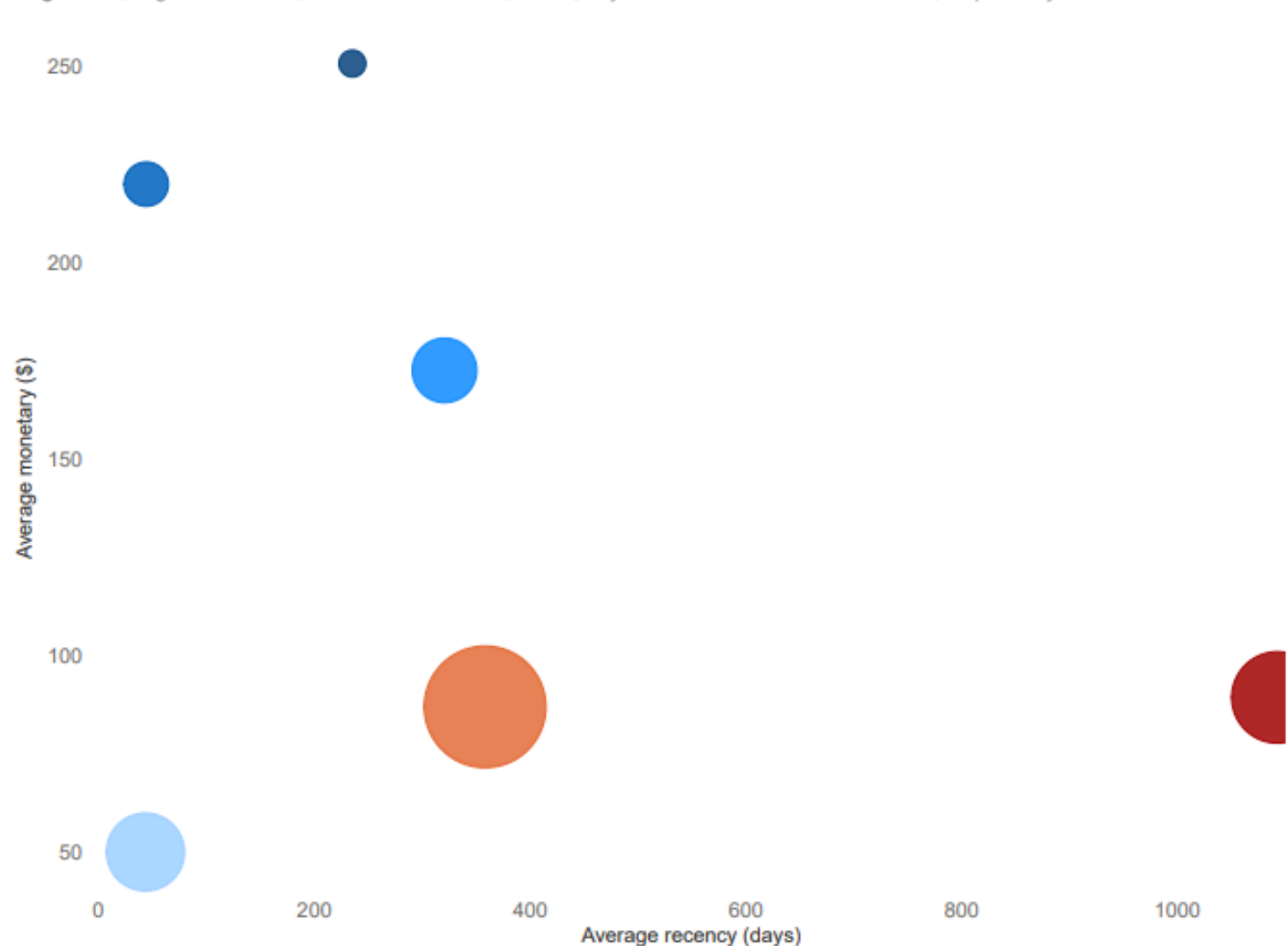
*High-Value New customers place the largest, highest-value baskets*

## Segment Performance Summary

| segment            | Complete Orders | Num Items (Complete) | AOV    | Items per Order | Orders per Customer |
|--------------------|-----------------|----------------------|--------|-----------------|---------------------|
| High-Value New     | 1215            | 2657                 | 219.89 | 2.19            | 1.00                |
| Inactive Customers | 12152           | 17650                | 86.85  | 1.45            | 1.00                |
| Lost               | 6529            | 9452                 | 89.25  | 1.45            | 1.00                |
| Loyal Customers    | 997             | 1413                 | 81.93  | 1.42            | 3.06                |
| New Customers      | 4571            | 5621                 | 49.90  | 1.23            | 1.00                |
| Repeat Buyers      | 5964            | 8629                 | 86.25  | 1.45            | 2.00                |

## RFM Positioning by Customer Segment

Segment ● High-Value New ● Inactive Customers ● Lost ● Loyal Customers ● New Customers ● Repeat Buyers



## Takeaways:

1. Loyal Customers generate value through frequency, averaging 3.06 orders per customer despite a modest \$81.93 AOV.
2. High-Value New Customers generate value through larger orders, with the highest AOV (\$219.89) and basket size (2.19 items/order) despite purchasing only once.
3. Repeat Buyers show a potential path toward loyalty, averaging 2.00 orders per customer with an AOV of \$86.25.
4. Inactive, Lost, and New Customers remain one-time buyers, averaging just 1.00 order per customer, with New Customers having the lowest AOV at \$49.90.

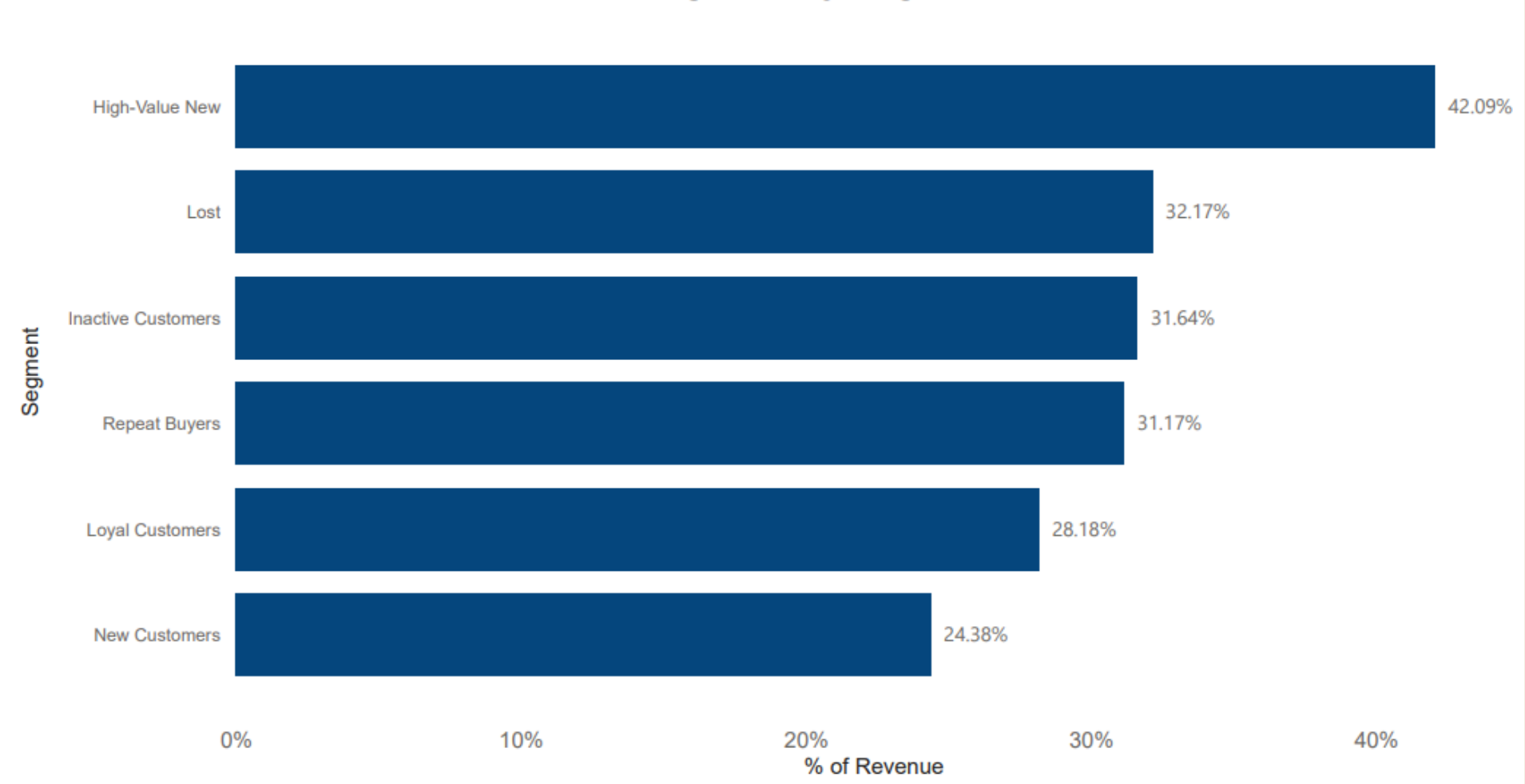
# PRODUCT PREFERENCE

*High-Value New customers concentrate 42.09% of spend in three categories*

Top 5 Product Categories by Segment

| category_rank | High-Value New                 | Inactive Customers                       | Lost                           | Loyal Customers               | New Customers                            | Repeat Buyers                            |
|---------------|--------------------------------|------------------------------------------|--------------------------------|-------------------------------|------------------------------------------|------------------------------------------|
| 1             | Outerwear & Coats (\$51,018)   | Jeans (\$129,585)                        | Outerwear & Coats (\$71,751)   | Jeans (\$9,396)               | Jeans (\$19,316)                         | Outerwear & Coats (\$62,008)             |
| 2             | Jeans (\$38,621)               | Outerwear & Coats (\$123,447)            | Jeans (\$71,451)               | Outerwear & Coats (\$7,478)   | Fashion Hoodies & Sweatshirts (\$19,108) | Jeans (\$59,686)                         |
| 3             | Sweaters (\$22,797)            | Sweaters (\$80,888)                      | Sweaters (\$44,237)            | Sweaters (\$6,146)            | Swim (\$17,191)                          | Sweaters (\$38,638)                      |
| 4             | Suits & Sport Coats (\$20,369) | Fashion Hoodies & Sweatshirts (\$68,316) | Suits & Sport Coats (\$37,665) | Swim (\$6,003)                | Intimates (\$16,387)                     | Suits & Sport Coats (\$31,950)           |
| 5             | Dresses (\$15,481)             | Swim (\$62,102)                          | Swim (\$36,941)                | Suits & Sport Coats (\$5,762) | Sweaters (\$16,132)                      | Fashion Hoodies & Sweatshirts (\$31,186) |

Proportion of Revenue from Top 3 Categories by Segment



## Takeaways:

High-Value New customers have the highest top-three category revenue concentration, while New Customers have the lowest.

- 1.High-Value New have the highest top-3 revenue concentration (42.09%), led by Outerwear & Coats (\$51,018).
  - 2.New Customers: Top-three concentration is the lowest at 24.38%, indicating the lowest revenue concentration in the segment’s top three categories.
  - 3.Jeans, Outerwear & Coats, and Sweaters rank among the top categories across most segments, making them core categories rather than segment-specific preferences.
- Overall, the interesting distinction is not simply what customers buy, but how concentrated their spending is across categories.

# CUSTOMER SEGMENTATION

*Loyal customers place more orders with similar basket sizes to Repeat Buyers*

## Segment Performance Comparison

| segment         | Complete Orders | Num Items (Complete) | AOV   | Items per Order | Orders per Customer |
|-----------------|-----------------|----------------------|-------|-----------------|---------------------|
| Loyal Customers | 997             | 1413                 | 81.93 | 1.42            | 3.06                |
| Repeat Buyers   | 5964            | 8629                 | 86.25 | 1.45            | 2.00                |

### Takeaways:

- 1.AOV is nearly identical (\$86.25 vs. \$81.93) and basket size is nearly identical (1.45 vs. 1.42 items/order) – meaning Loyal Customers don't spend more per order than Repeat Buyers.
  - 2.The real gap is purchase frequency: Loyal Customers average 3.06 orders/customer vs. 2.00 for Repeat Buyers – a difference of ~1 order.
  - 3.Repeat Buyers form a larger customer group than Loyal Customers (2,982 vs. 326), providing a broader base for initiatives encouraging an additional purchase.
- Similar AOV and items per order suggest that encouraging additional purchases is a reasonable focus for Repeat Buyers.
- A frequency-based loyalty incentive is one potential approach.

Capstone Project – Track 2

# THANK YOU

Thanks For Watching

Presented by Nguyen Thi Van Anh